

GEOGRAPHY

PHYSICAL GEOGRAPHY



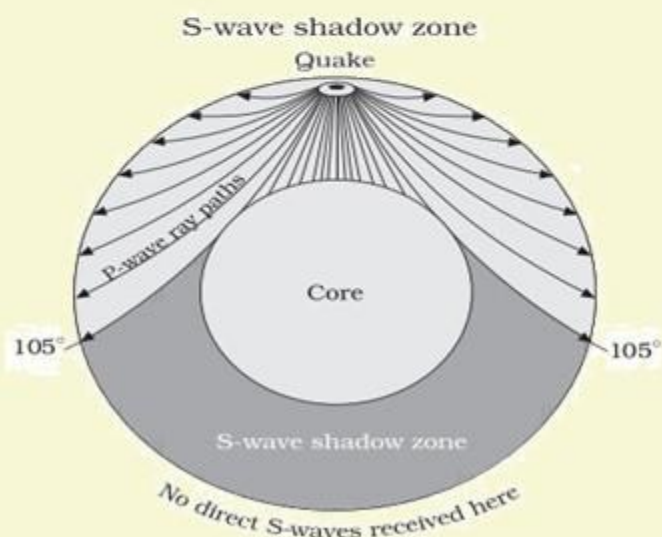
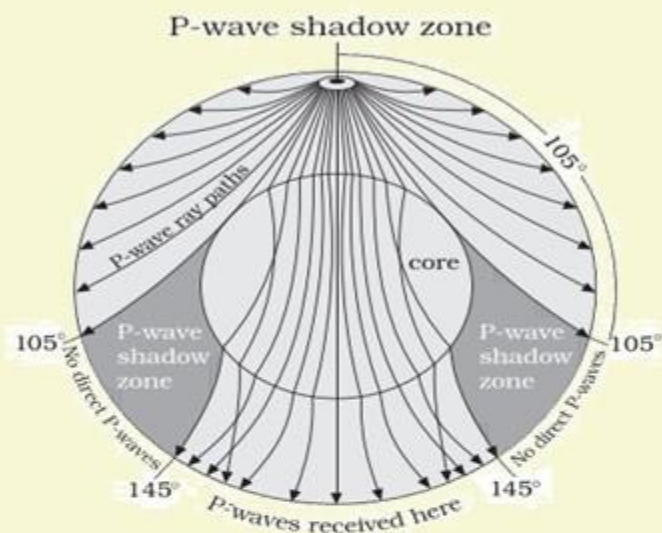
GEOMORPHOLOGY

MODULE - 24

EARTHQUAKES

- SHADOW ZONE OF P AND S WAVES
- MEASURING AN EARTHQUAKE

EMERGENCE OF SHADOW ZONE

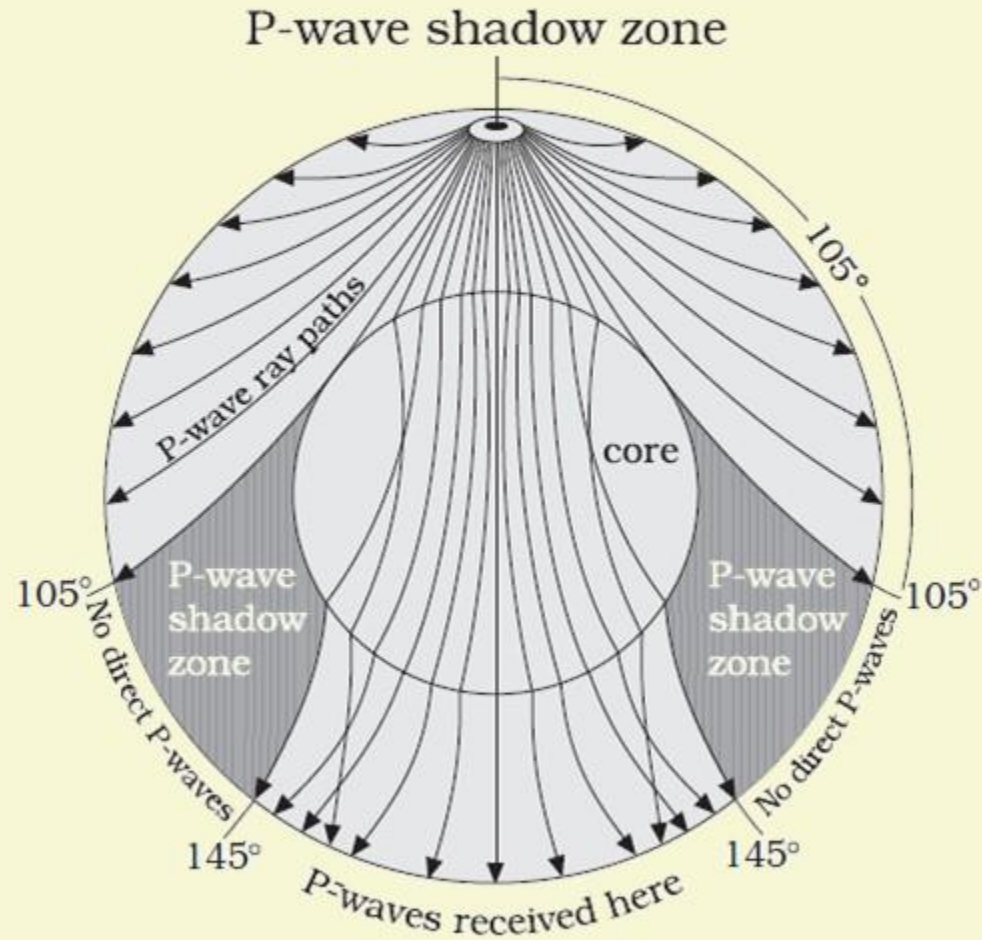


1 Earthquake waves get recorded in seismographs located at far of locations. However, there exist some specific areas, where the waves are not reported. Such a zone is called the “Shadow Zone”.

2 The study of different events reveals that for each Earthquake there exists an altogether different shadow zone.

EMERGENCE OF SHADOW ZONE

SHADOW ZONE OF P-WAVES



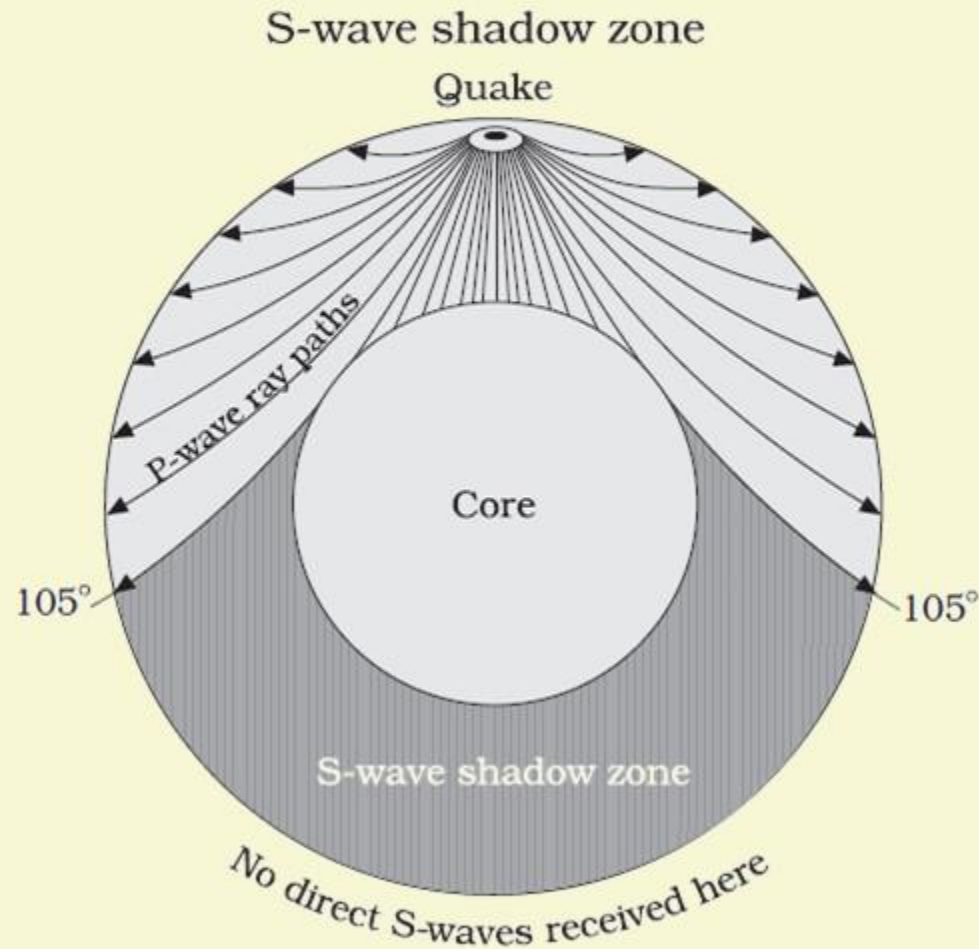
3 It was observed that seismographs located at any distance of 105° from the epicentre, record the arrival of both P and S-waves.

4 However, the Seismographs located beyond 145° from epicentre, record the arrival of P-waves, but not that of S-waves.

5 Thus, a zone between 105° and 145° from epicentre was identified as the shadow zone for both the types of waves.

EMERGENCE OF SHADOW ZONE

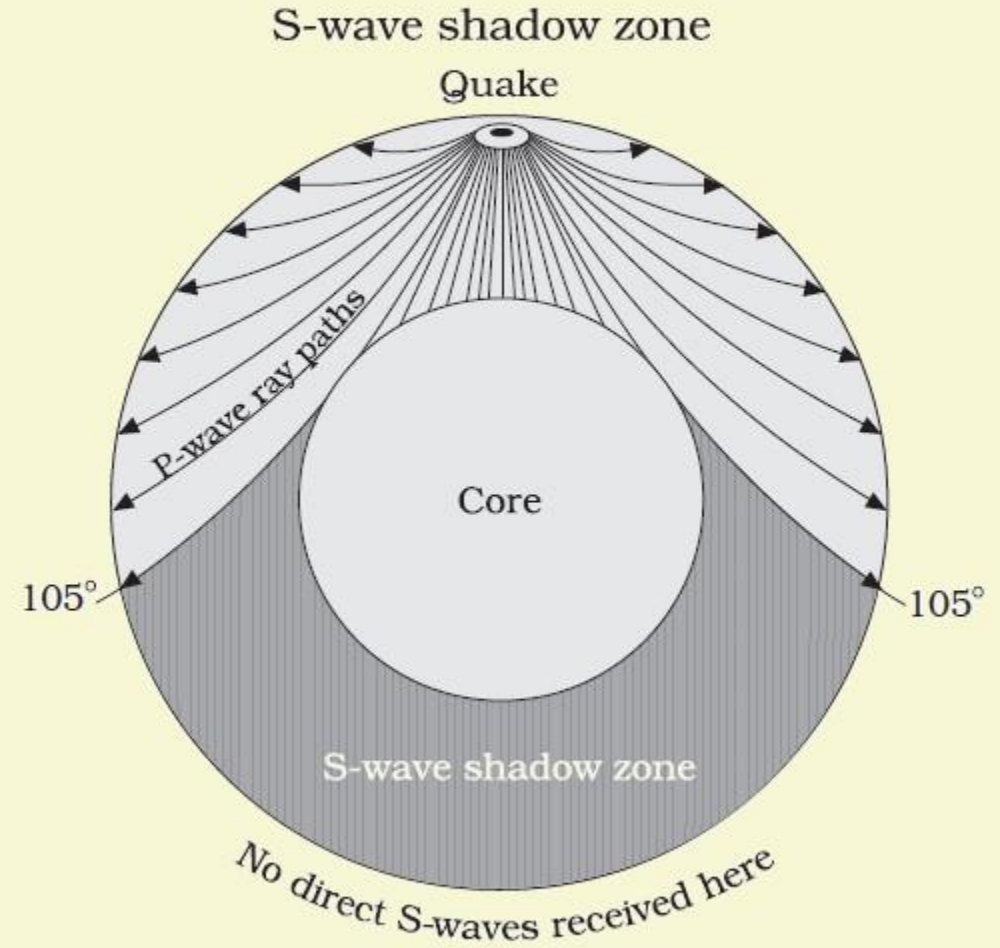
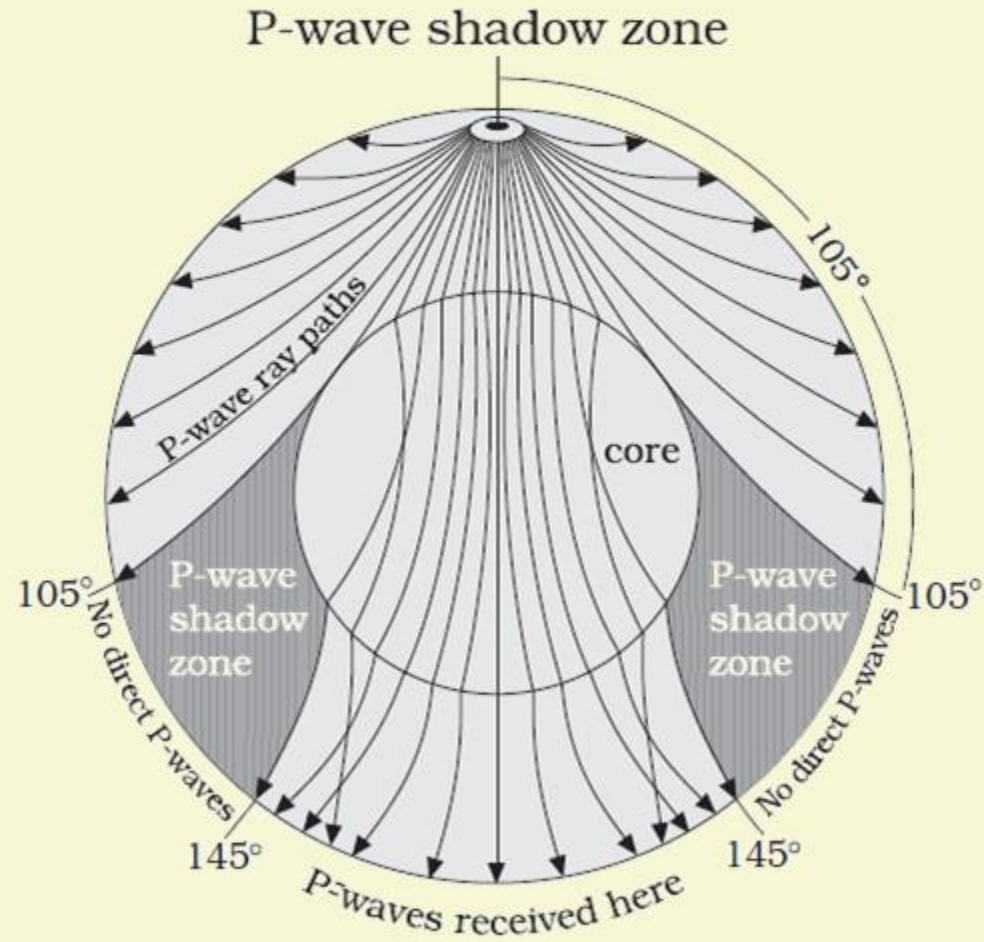
SHADOW ZONE OF S-WAVES



- 6 The entire zone beyond 105° does not receive S-waves. The shadow zone of S-wave is much larger than that of the P-waves.
- 7 The shadow zone of P-waves appears as a band around the Earth between 105° and 145° away from the epicentre.
- 8 The shadow zone of S-waves is not only larger in extent but it is also a little over 40 percent of the Earth Surface.

EMERGENCE OF SHADOW ZONE

SHADOW ZONE OF P AND S-WAVES



TYPES OF EARTHQUAKES

Tectonic Quake



Volcanic Quake



1 The most common ones are the **tectonic Earthquakes**. These are generated due to **sliding of rocks along a fault plane**.

2 A special class of tectonic Earthquake is sometimes recognized as **volcanic Earthquake**. However, these are confined to areas of active volcanoes.

3 In the areas of **intense mining activity**, sometimes the roofs of underground mines collapse causing **minor tremors**. These are called **collapse Earthquakes**.

TYPES OF EARTHQUAKES

- 4 Ground shaking may also occur due to the explosion of chemical or nuclear devices. Such tremors are called **explosion Earthquakes**.



- 5 The Earthquakes that occur in the areas of large reservoirs are referred to as **reservoir induced Earthquakes**.



MOCK QUESTION

Which of the following can cause an Earthquake?

- 1) Volcanic Explosion
- 2) Active Fault zones
- 3) Tsunami waves
- 4) Mining
- 5) Construction of dams and reservoirs

Choose the correct statement from the ones given below

- | | |
|----------------|---------------------|
| a) 1, 2, 3 & 4 | b) 1, 2, 4, & 5 |
| c) 1, 2 & 3 | d) All of the above |

ANSWER: B

EXPLANATION



1. Tsunami waves are the result of earthquakes and they DO NOT cause them.
2. **RIS : Reservoir Induced Seismicity**
Hydrostatic pressure of human constructed dams and reservoirs cause **disequilibrium** of isostatically adjusted rocks and produce **earthquakes**. This is one of the important anthropogenic causes of earthquake.
E.g.: Koyna earthquake of 1967, in Satara district of Maharashtra due to Koyna reservoir.

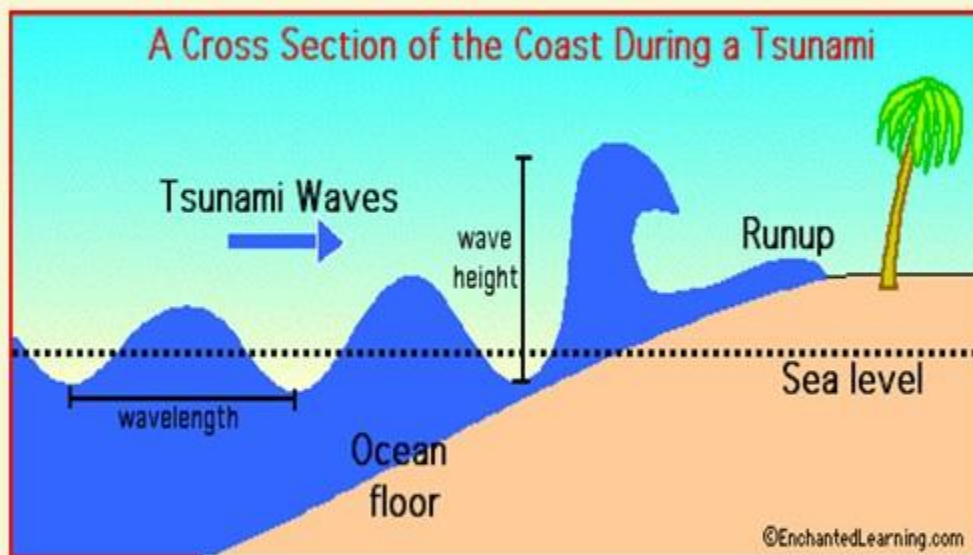
MOCK QUESTION

With reference to Tsunamis, consider the following statements

- 1) Tsunamis are only generated by undersea earthquakes and volcanic eruptions.
- 2) As it approaches the coast, its velocity increases

Choose the correct statement

- | | |
|---------------|--------------------|
| a) 1 only | b) 2 only |
| c) Both 1 & 2 | d) Neither 1 nor 2 |



ANSWER: D

EXPLANATION

1. Generated by undersea earthquakes, volcanic eruption, displacement of seawater due to landslides, fall of meteorites in the ocean.
2. As it approaches the coast, the waves slow down in velocity, but its amplitude increases leading to huge destruction along the shore.

MEASURING EARTHQUAKES



GIUSEPPE MERCALLI

- 1 The Earthquake events are scaled either according to the **magnitude or intensity of the shock**.
- 2 The magnitude scale is known as the **Richter scale**.
- 3 The magnitude relates to the energy released along the fault during the quake.
- 4 The Richter scale is expressed in absolute numbers 0-10.
- 5 The **Intensity scale** is named after Mercalli, an **Italian seismologist**. The intensity scale takes into account the visible damage caused by the event. The range of intensity scale is from 1-12.

DIFFERENCE BETWEEN THE RICHTER AND MERCALLI SCALE

RICHTER SCALE	MERCALLI SCALE
The Richter scale describes the earthquake's magnitude by measuring the seismic waves that cause the earthquake.	The Mercalli scale quantifies the effect an earthquake has on humans , natural objects and man-made structures based on observation.
Obtained by calculating logarithm of the amplitude of waves.	Quantified from observation of effect on earth's surface, human, objects and man-made structures
Base-10 logarithmic scale i.e. a 4.0 earthquake is 10 times stronger than a 3.0 earthquake	I (not felt) to XII (total destruction)

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